

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				1-2 marks Description of reactions only or fair attempt at equations <i>The candidate attempts to link at least two relevant points from the indicative content. Coherence is limited by omission and/or inclusion of irrelevant material. There is some evidence of appropriate use of scientific conventions and vocabulary.</i> 0 marks <i>The candidate does not make any attempt or give an answer worthy of credit.</i>						
	(b)			$n(\text{CO}_2) = \frac{186}{24500} = 7.59 \times 10^{-3} \quad (1)$ $n(\text{K}_2\text{CO}_3) = 7.59 \times 10^{-3}$ $\text{mass K}_2\text{CO}_3 = 7.59 \times 10^{-3} \times 138.2 = 1.05 \text{ g} \quad (1)$ $\text{mass impurity} = 1.40 - 1.05 = 0.35 \text{ g} \quad (1)$		2	1	3	2	
	(c)			flame test - lilac flame (1) (HNO_3 followed by) $\text{AgNO}_3(\text{aq})$ - white precipitate (1)	2			2		2
	(d)			large increase from 1st to 2nd ionisation energy / high 2nd ionisation energy / too much energy needed to remove 2nd electron (1) second electron removed from a shell nearer to the nucleus / with increased effective nuclear attraction / less shielding (1)	2			2		
				Question 10 total	8	4	1	13	2	3